Cybersecurity: How Blockchains can Help

Blockchains offer a great platform for organisations and governments to secure their digital assets in a tamper-proof way.

Cybersecurity refers to the mechanisms that protect digital infrastructure, networks and programs from malicious attacks and unauthorised access. Security controls are put in place to safeguard sensitive information and ensure confidentiality, integrity and availability of data.

Malware (such as viruses, worms, or Trojan horses) can disrupt systems by corrupting files or stealing sensitive information. Phishing attacks use fake emails or websites to deceive users into providing their login credentials or financial information. Denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks flood servers with so much traffic that websites and software applications become unavailable. In a man-in-the-middle attack, a cyber criminal intercepts the communication between two parties to steal information or manipulate a message. Zero day exploits take advantage of the unknown vulnerabilities of the software so that hackers can infiltrate a network before a software patch is released.

In the era of artificial intelligence, deepfake technology is adding to the complexity of the cyber security landscape. Another key risk is insider attacks, where employees or partners misuse their access to compromise data. While the number of IoT (Internet of Things) devices being used is steadily increasing, so are the IoT security threats; connected devices often have poor security protocols and are susceptible to hacking.

Businesses try and counteract these threats with firewalls, intrusion detection systems, encryption, multi-factor authentication, and advanced endpoint security solutions. There are multiple cyber security frameworks in place (NIST, ISO/IEC 27001, CIS Controls, etc) to help protect digital assets. Continuous threat intelligence, security audits and employee training are key to a strong cyber security defence designed to detect and prevent specific attacks.

The changing landscape of cyber threats has made proactive security strategies increasingly important, whether using blockchain technology to secure transactions or AI-driven threat detection to protect the digital ecosystem.

Figure 1: Cybersecurity and digital forensics

Using blockchain for cybersecurity

Blockchain technology is redefining cybersecurity by offering decentralised, immutable and transparent solutions for different security issues. One of the key applications of blockchain in cybersecurity is identity management, as it provides secure and tamper-proof identities that lower the chance of identity theft and fraud.

The use of decentralised storage solutions based on blockchain makes it virtually impossible for hackers to modify or steal crucial data. Blockchains can also help secure IoT devices and transmitted data by allowing authenticated devices to share data over secure ledgers.

The blockchain environment enhances incident response and threat intelligence sharing by providing a